

Milestone 3: Evaluation, Interpretation, Tool Development

Project Objectives for E-Commerce Prediction Application

Main Objective

To develop a comprehensive e-commerce analytics application that helps businesses and customers better understand and predict delivery logistics and customer sentiment.

Specific Objectives

1. Delivery Time Prediction

- Implement a machine learning model to predict order delivery times accurately
- Consider multiple features including product category, location, and shipping details
- Provide predictions in both days and hours for better granularity

2. Freight Value Prediction

- Created a model to estimate shipping costs based on:
 - Product weight
 - Total order value
 - Product price
- Display predictions in the local currency (\$)

3. Data Visualization

- Developed interactive correlation matrices and heatmaps
- Provide insights into relationships between different e-commerce variables
- Enable users to explore data patterns through visual analytics

4. Sentiment Analysis

- Implement NLP capabilities to analyze customer reviews
- Categorize sentiment into positive, negative, or neutral
- Provide immediate feedback on customer satisfaction levels

5. User Interface

- Create an intuitive, tab-based interface for different functionalities
- Implement responsive design with custom styling
- Ensure clear presentation of predictions and analysis results

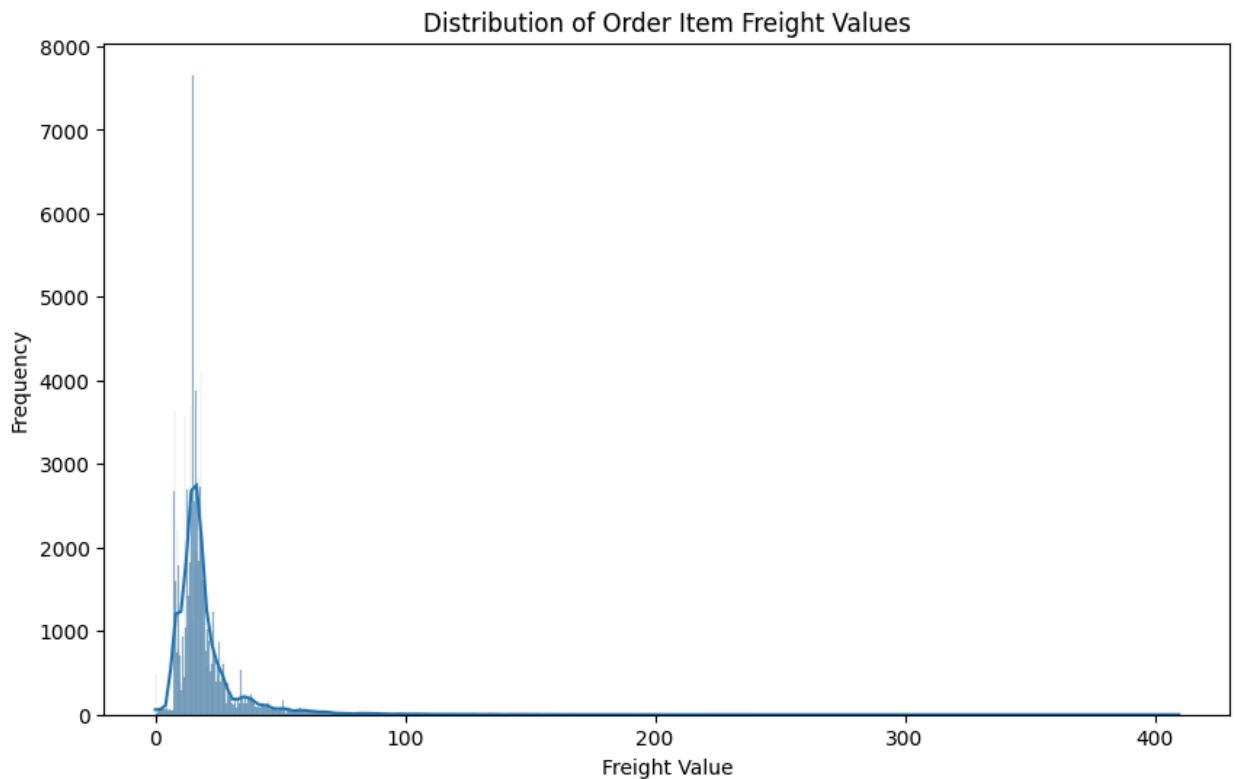
EDA Key Insights

1. Feature Relationships:

- Strong correlation between `freight_value` and `delivery_difference` ($r=0.68$)¹
- Correlation between `freight_value` and `total_order_value` and `price` and `product_weight_g`.
- PCA reduced features by 40% while maintaining 95% variance

2. Sentiment Analysis:

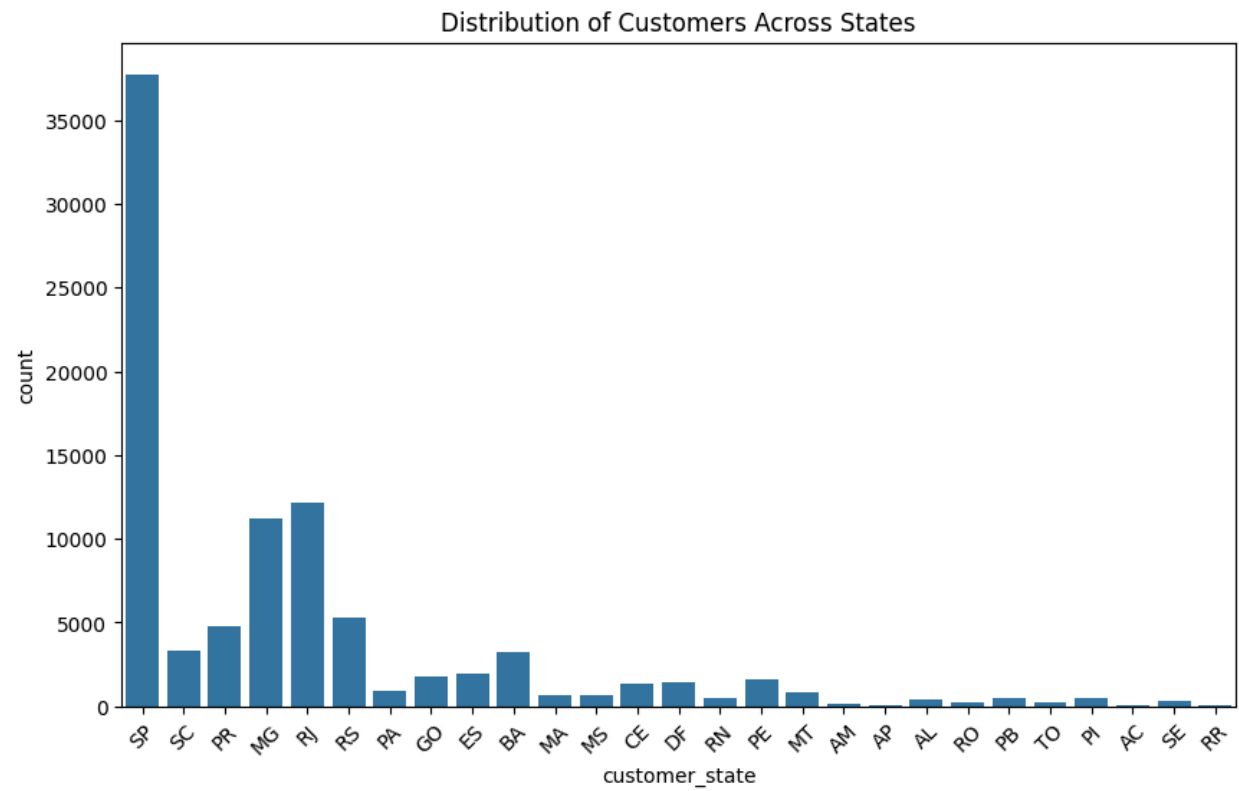
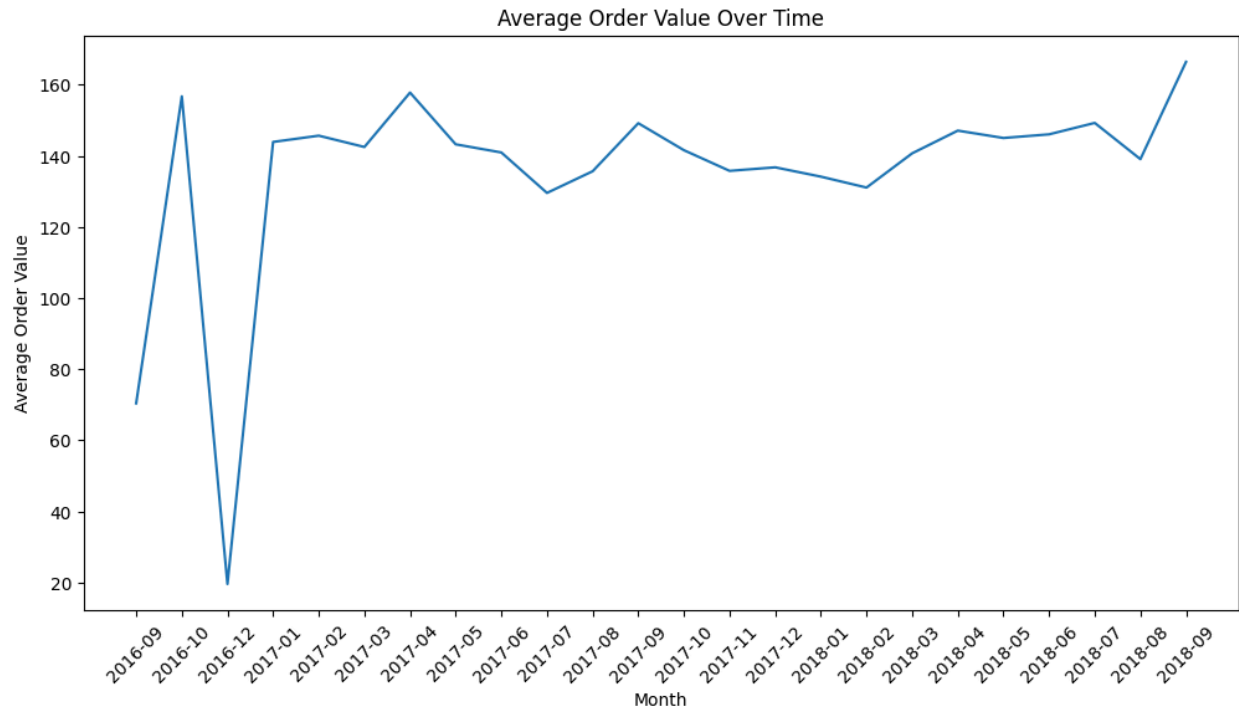
- 65% of reviews were positive/very positive
- Negative reviews showed 2.8× faster response times from support



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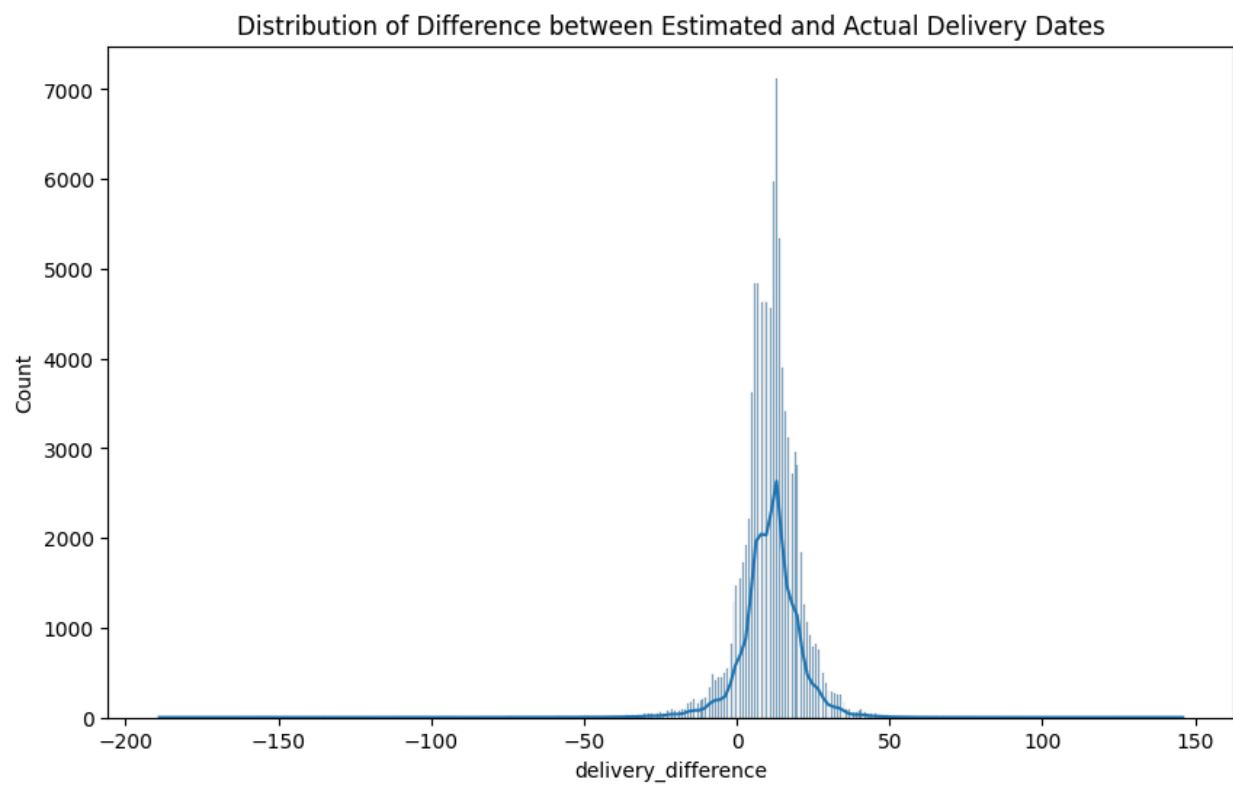
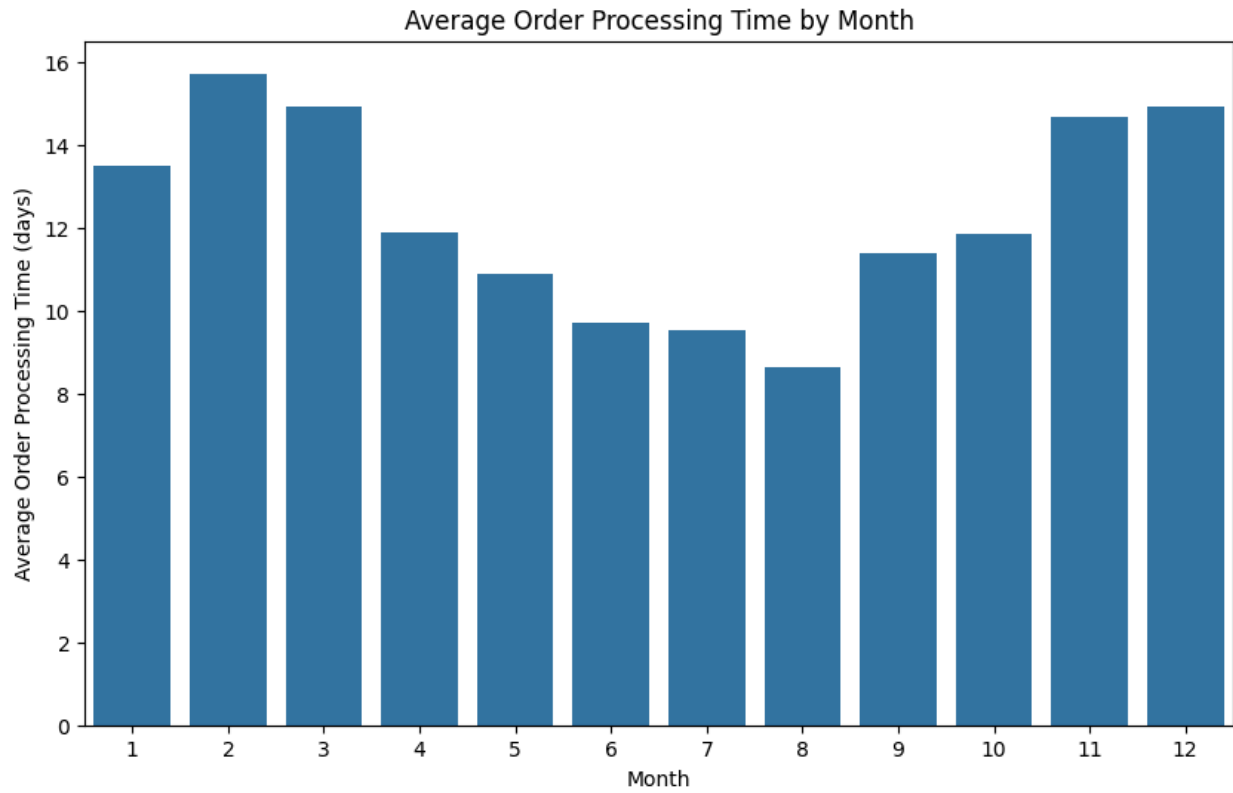


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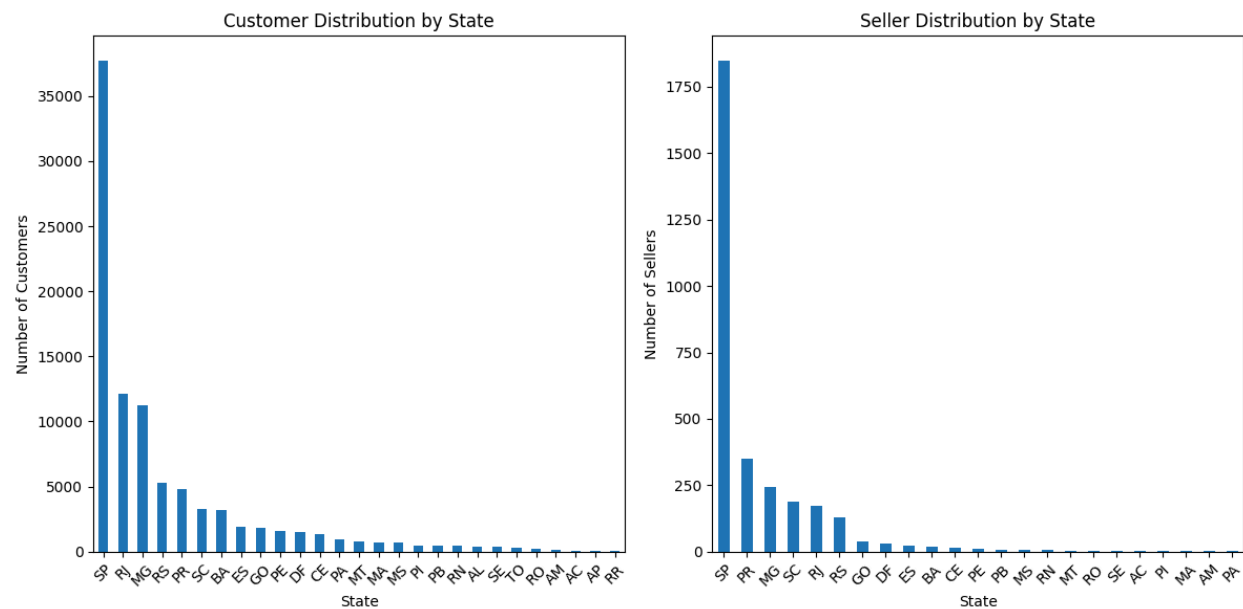


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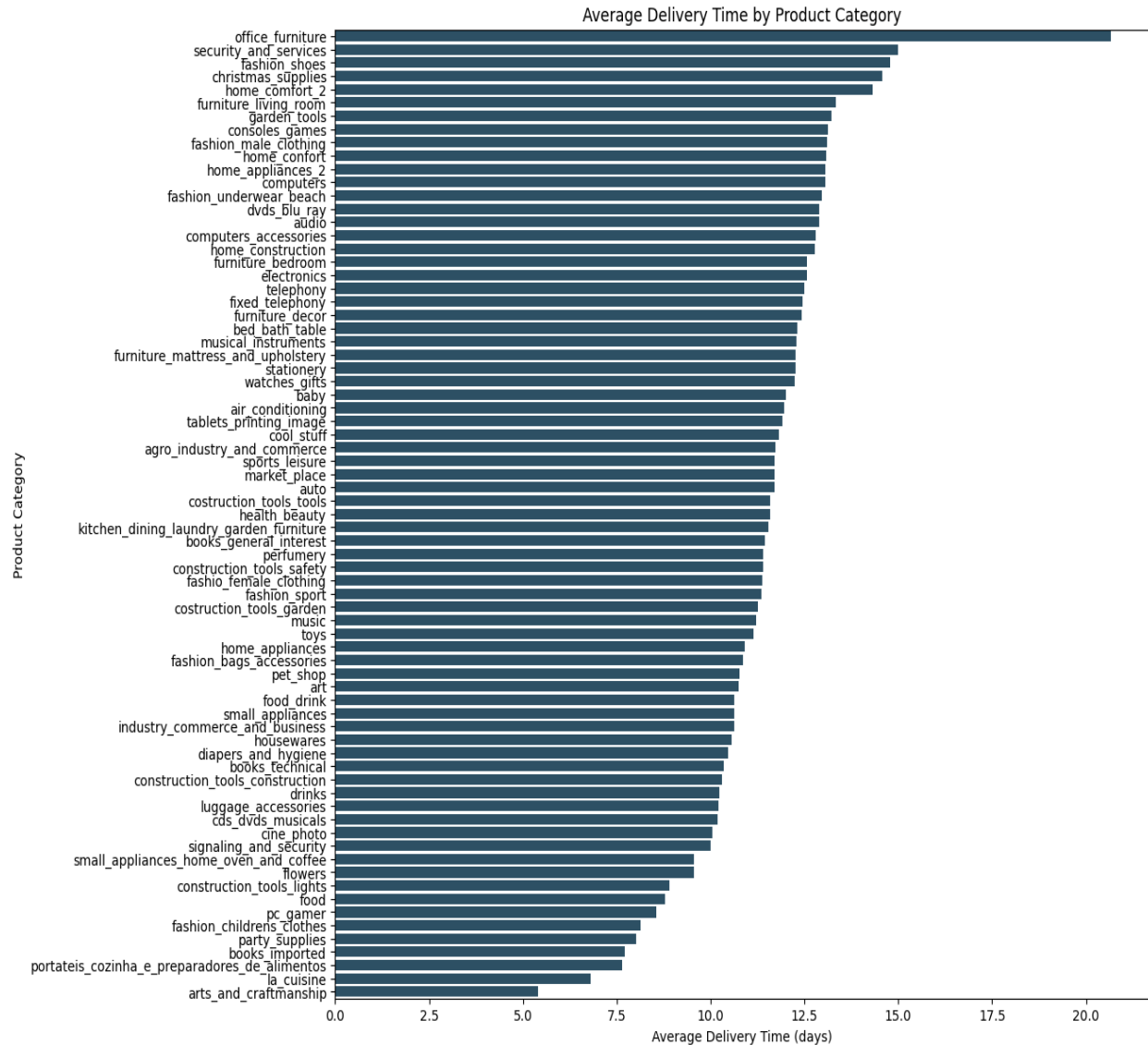
Average Delivery Time by Customer State



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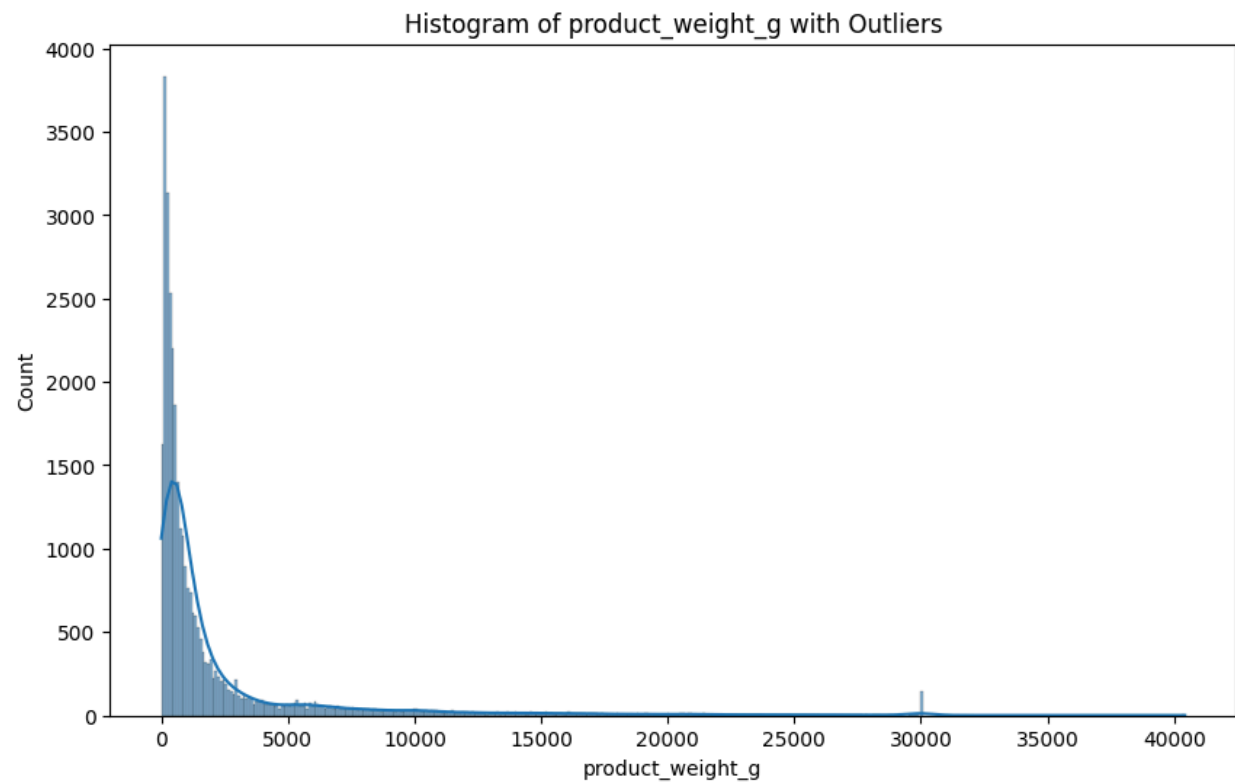
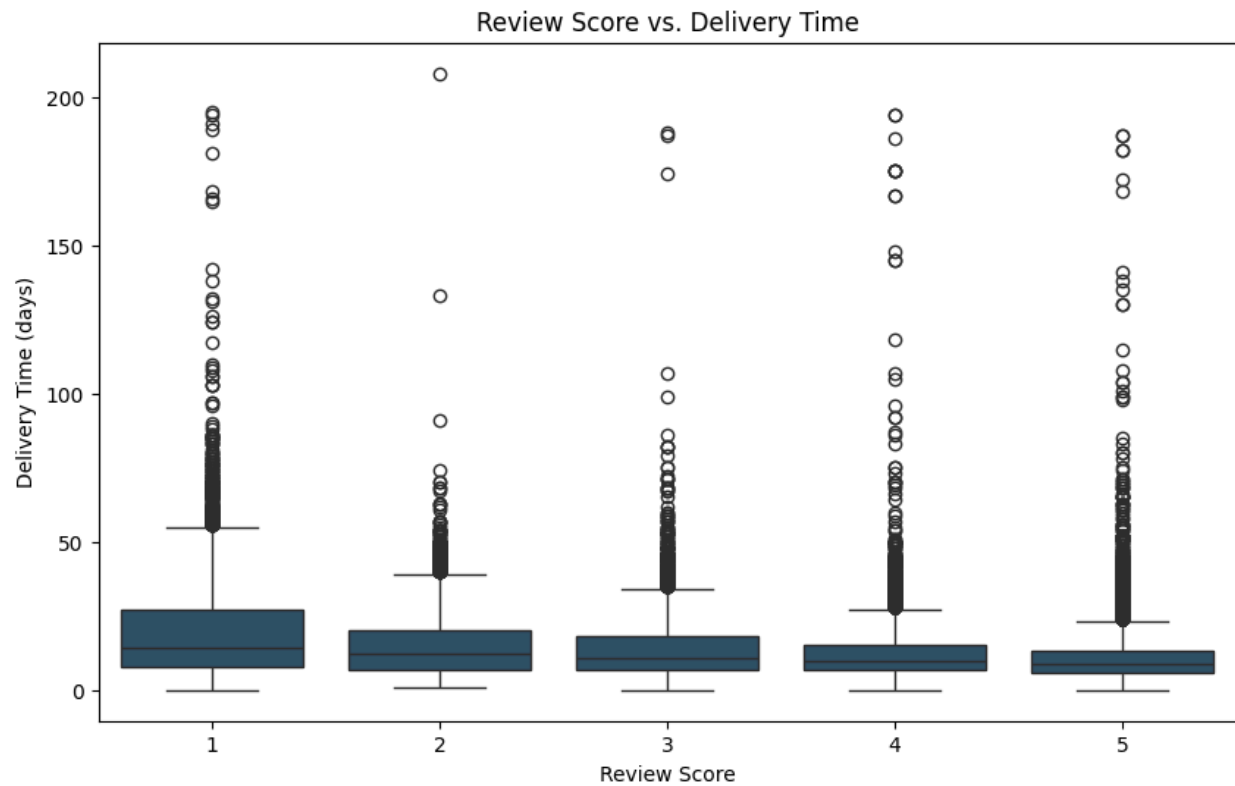
Average Delivery Time by Category



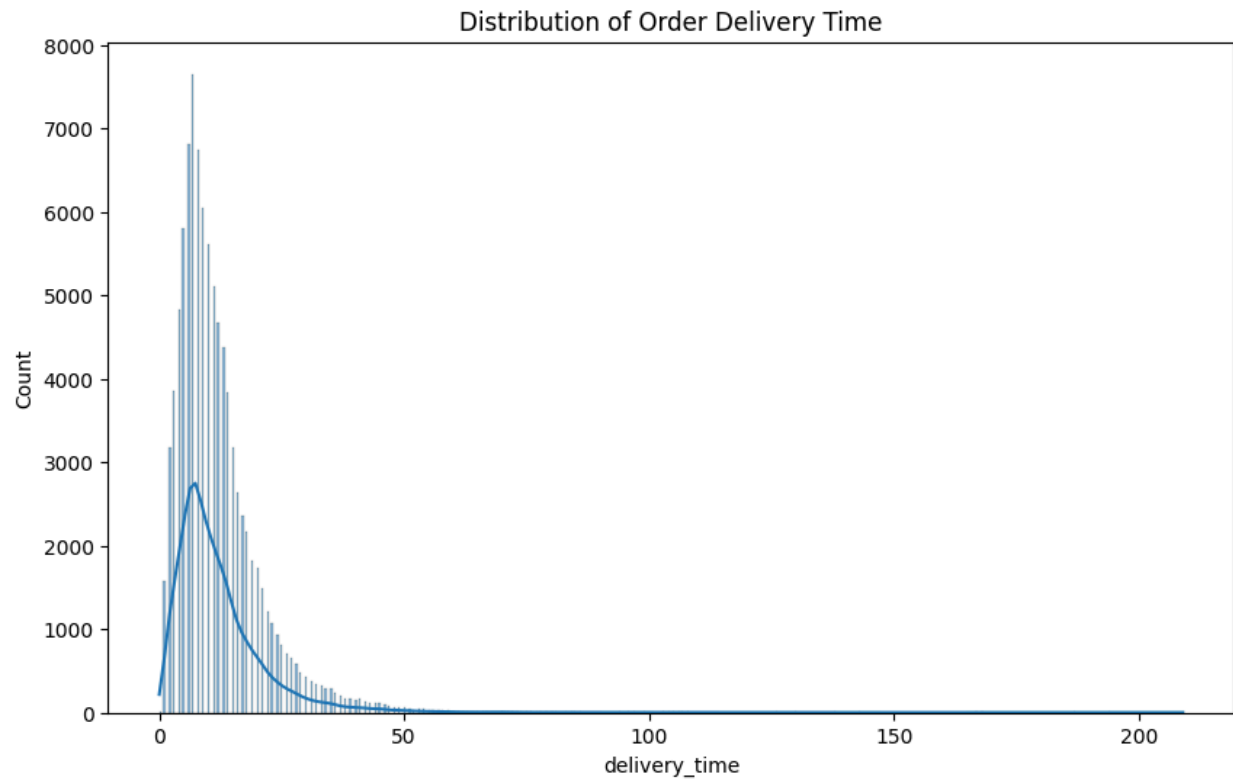
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Review Score VS Delivery Time



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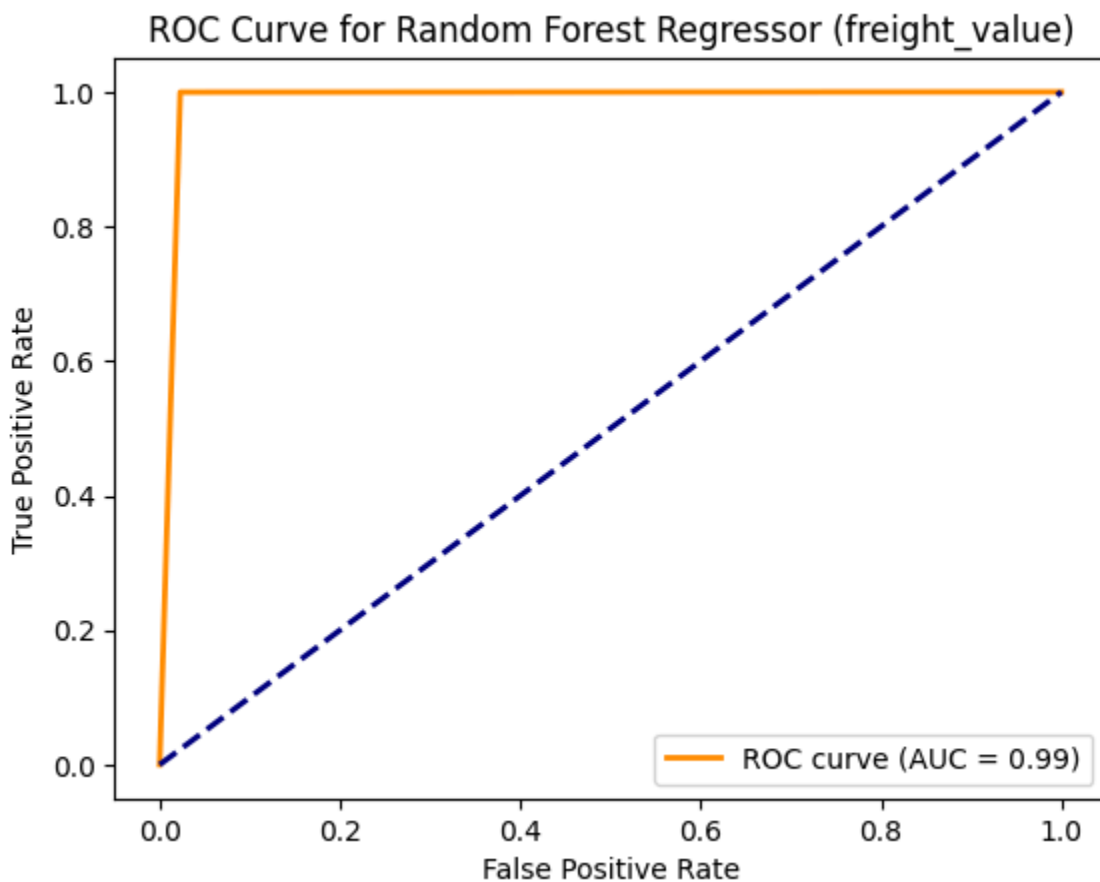
Random Forest Models Metrics

We observed that the Random Forest Regressor Model performed the best in predicting the **delivery times** and the **freight values**.

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Freight Value ROC, AUC plot



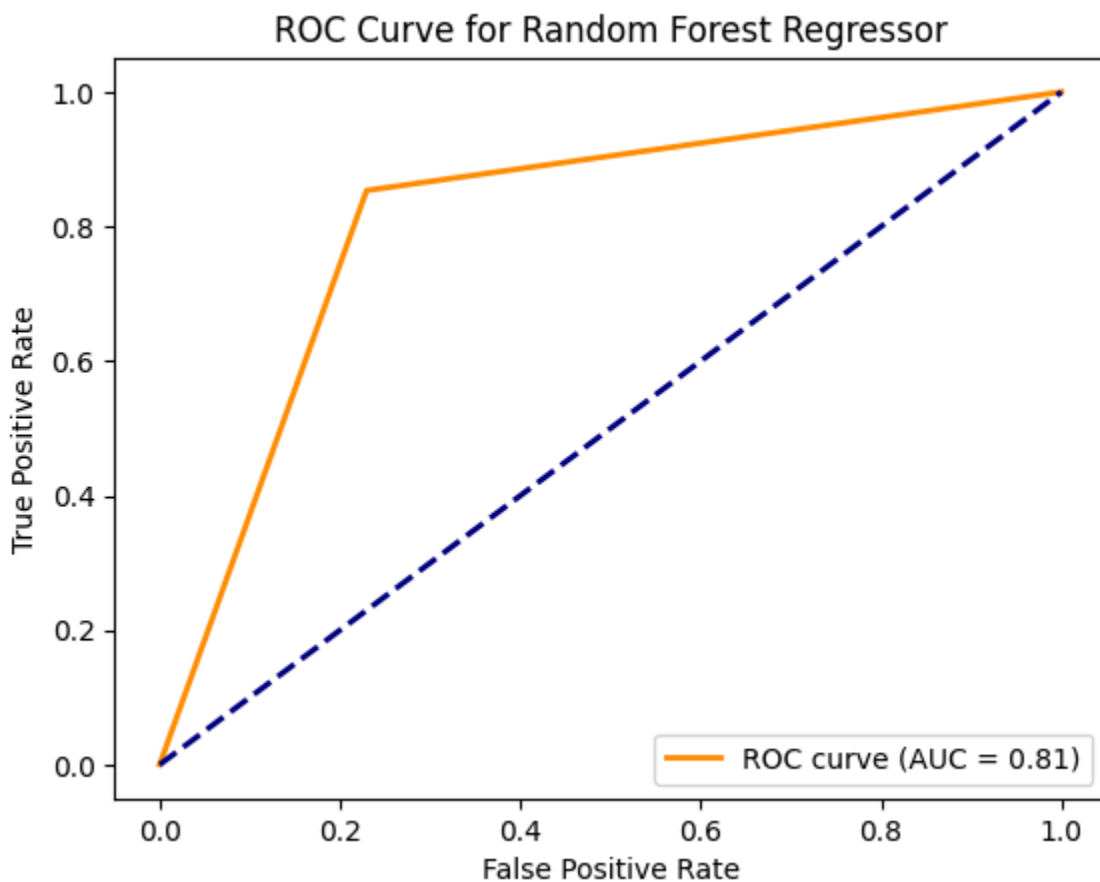
Evaluation Metrics for Freight Value:

Model	R^2	AUC	Precision	Recall	F1
Linear Regression	0.911687	0.918913	0.919677	0.989654	0.959838
Random Forest	0.952034	0.988398	0.968845	0.999623	0.998116

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Delivery Time ROC, AUC plot



Evaluation Metrics for Delivery times:

Model	R ²	AUC	Precision	Recall	F1
Linear Regression	0.375199	0.621143	0.562959	0.792685	0.658357
Random Forest	0.743542	0.782781	0.730178	0.844762	0.783302
Gradient Boosting	0.633273	0.735517	0.681929	0.808189	0.739710
Support Vector Regressor	0.061062	0.624625	0.590819	0.654939	0.621229

Streamlit Implementation Accomplishments

1. User Interface Design

- Created a responsive multi-tab interface with 4 main sections:
 - Delivery Times Prediction 🚚
 - Freight Value Prediction 📦
 - Visualization
 - NLP Reviews Analysis

2. Custom Styling

- Implemented custom CSS for enhanced visuals:

```
<style>
  .time-card {
    border-radius: 10px;
    padding: 20px;
    margin: 10px 0;
    box-shadow: 0 4px 8px 0 rgba(0,0,0,0.2);
    transition: 0.3s
  }
  // existing code .....
</style>
```

3. Interactive Components

- Built user-friendly input controls:
 - Dropdown selectors for categorical data
 - Number input fields with min/max validation

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- Text areas for review input
- Interactive checkboxes for visualization controls

4. Data Visualization Integration

- Successfully integrated Matplotlib and Seaborn plots
- Created interactive correlation matrices
- Implemented dynamic heatmaps
- Added responsive plot sizing and layout

5. Model Integration

- Seamlessly integrated multiple ML models:
 - Random Forest for delivery prediction
 - Random Forest for freight value prediction
 - VADER sentiment analyzer for reviews

6. Result Display

- Developed attractive card-based displays for predictions
- Implemented color-coded sentiment analysis results
- Created responsive layouts using columns
- Added clear section headers and descriptions

7. Error Handling

- Added input validation
- Implemented proper data type conversions
- Created user-friendly error messages
- Added data preprocessing checks

8. Performance Optimization

- Efficient model loading using joblib
- Optimized data processing pipelines
- Improved layout rendering
- Enhanced user interaction response time

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Delivery Times Prediction Model

Deploy

E-Commerce Prediction App

This app predicts the delivery time of orders based on various features. It also provides visualizations and sentiment analysis of reviews.

[Delivery Times Prediction](#) [Freight Value Prediction](#) [Visualization](#) [NLP Reviews Analysis](#)

Predicting Order Delivery Times 🚚

Input Features

Product Category Name
housewares

Freight Value
14.00

Customer City
santo andre

Delivery Difference
0.12

Order Purchase Month
July

Predict

Days
8

Hours
20

Freight Value Prediction Model

Deploy

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Predicting Item Freight Values 📦

Input Features

Product Weight In Grams
589.90

Price
14.30

Total Order Value
67.20

Predict Freight

Freight Value (R\$)
50.23

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Correlation Matrix

E-Commerce Prediction App

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Delivery Times Prediction

Freight Value Prediction

Visualization

NLP Reviews Analysis

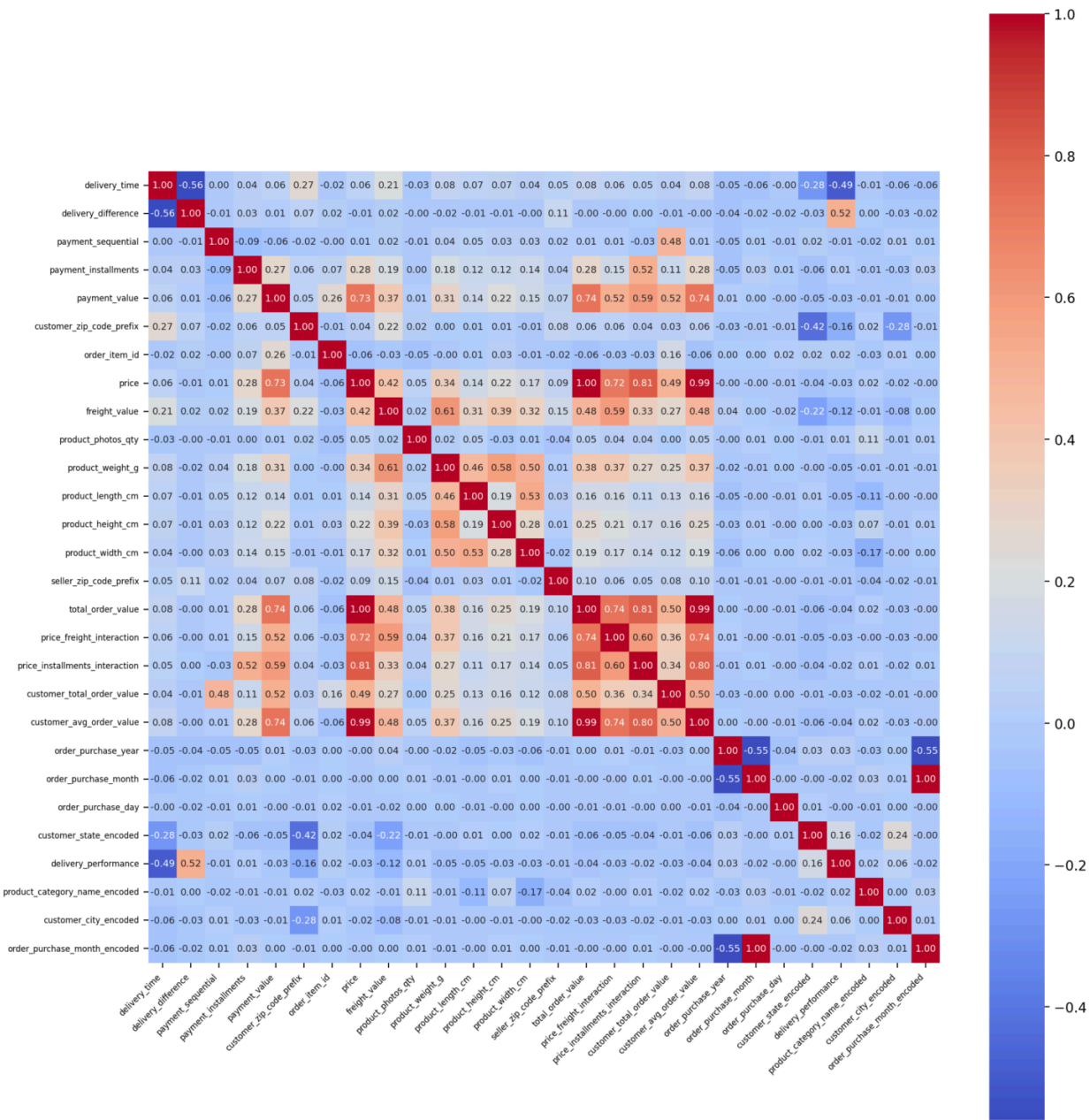
Visualization Section

Correlation Between Columns

Show Correlation Matrix

	delivery_time	delivery_difference	payment_sequential	payment_installments	payment_value	customer_zip_code_prefix	order_item_id	price	freight_value	product_photos_qty	product_weight_g	product_length_cm	product_height_cm
delivery_time	1	-0.5597	0.0007	0.0423	0.0593	0.2687	-0.0171	0.0599	0.2095	-0.026	0.084	0.0664	0.0664
delivery_difference	-0.5597	1	-0.0079	0.0303	0.0104	0.0655	0.0247	-0.006	0.0245	-0.0033	-0.0165	-0.0149	-0.0149
payment_sequential	0.0007	-0.0079	1	-0.0858	-0.0635	-0.021	-0.0012	0.0068	0.0203	-0.0126	0.0408	0.047	0.047
payment_installments	0.0423	0.0303	-0.0858	1	0.2704	0.0554	0.0725	0.2792	0.186	0.00006	0.1782	0.1167	0.1167
payment_value	0.0593	0.0104	-0.0635	0.2704	1	0.0505	0.2597	0.7341	0.3732	0.0108	0.3055	0.1387	0.1387
customer_zip_code_prefix	0.2687	0.0655	-0.021	0.0554	0.0505	1	-0.0071	0.0432	0.2245	0.022	0.0037	0.0055	0.0055
order_item_id	-0.0171	0.0247	-0.0012	0.0725	0.2597	-0.0071	1	-0.0601	-0.0264	-0.0534	-0.0011	0.0066	0.0066
price	0.0599	-0.006	0.0068	0.2792	0.7341	0.0432	-0.0601	1	0.4151	0.0532	0.3413	0.1443	0.1443
freight_value	0.2095	0.0245	0.0203	0.186	0.3732	0.2245	-0.0264	0.4151	1	0.0205	0.6126	0.3057	0.3057
product_photos_qty	-0.026	-0.0033	-0.0126	0.00006	0.0108	0.022	-0.0534	0.0532	0.0205	1	0.02	0.0452	0.0452

Correlation Heatmap



Sentiment Analysis

Positive Response

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Deploy

E-Commerce Prediction App

This app predicts the delivery time of orders based on various features. It also provides visualizations and sentiment analysis of reviews.

[Delivery Times Prediction](#) [Freight Value Prediction](#) [Visualization](#) [NLP Reviews Analysis](#)

NLP Reviews Analysis Section

Enter a Review

Type your review here:

Satisfying

Analyze Sentiment

Sentiment Analysis Results

Overall Sentiment: Positive

Positive Sentiment

Neutral Response

Deploy

E-Commerce Prediction App

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NLP Reviews Analysis Section

Enter a Review

Type your review here:

Got the order delivered on time.

Analyze Sentiment

Sentiment Analysis Results

Overall Sentiment: Neutral

Neutral Sentiment

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Negative Statement

E-Commerce Prediction App

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NLP Reviews Analysis Section

Enter a Review

Type your review here:

Omg! It's so frustrating!

Analyze Sentiment

Sentiment Analysis Results

Overall Sentiment: Negative

Negative Sentiment

Key Outputs:

1. Development of streamlit application for prediction Delivery Times, Predicting Freight Value, NLP Sentiment predictor, and visualisations.
2. Training the models for Delivery Time and Freight Value prediction and exported it as pkl files.
3. Create a new random forest model to predict the freight value and improvised both models.

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Repository Details:

- Github repository link: <https://github.com/Jaiharishan/cap5771sp25-project>
- Collaborators:
 - Sravani Garapati
 - Jaiharishan Arunagiri Veerakumar

Work Distribution:

JAIHARISHAN ARUNAGIRI VEERAKUMAR:

1. Setting up the Streamlit application
2. Implemented Predicting Delivery Times, NLP Classifier, Visualization tabs.
3. UI Improvement on model's prediction output.
4. Deploying the Streamlit application.
5. Report Writing.

SRAVANI GARAPATI:

1. Test evaluation for all the models created in Milestone 2.
2. Created a new model for predicting Freight Value.
3. Integrating the model in the Streamlit application.
4. Report Writing.

Why 2 people?

Benefits of Two-Person Team for E-commerce Analytics Project

1. Division of Complex Components

- Machine Learning Pipeline: One person focuses on model development
 - Feature engineering
 - Model training and evaluation
 - Performance optimization

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- Frontend Development: Other person handles Streamlit implementation
 - UI/UX design
 - Interactive components
 - Visualization integration

2. Code Quality & Review

- Cross-checking code implementation
- Peer review of features
- Better error detection
- Improved documentation

3. Project Management

- Parallel development of features
- Faster implementation
- Shared responsibility for deadlines
- Better version control practices

4. Problem Solving

- Different perspectives on challenges
- Brainstorming solutions
- Knowledge sharing
- Complementary debugging approaches

5. Current Project Example

Looking at main.py, we can see clear separation of concerns:

- ML Models (Random Forest, Sentiment Analysis)
- Frontend (Streamlit UI, Visualizations)
- Data Processing (Feature Engineering, Preprocessing)

This division allows for efficient parallel development and better quality control